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**Assignment 9**
**Consider the following tables namely “DEPARTMENTS” and “EMPLOYEES” Their schemas**
**are as follows,**
**Departments ( dept no , dept name , dept_location ); Employees ( emp_id , emp_name , emp_salary,dept_no);**

**a) Develop a query to grant all privileges of employees table into departments table**

```
GRANT ALL PRIVILEGES
ON Employees
TO 'dept_user';
```

**b) Develop a query to grant some privileges of employees table into departments table**

```
GRANT SELECT, INSERT
ON Employees
TO 'dept_user';
```

**c) Develop a query to revoke all privileges of employees table from departments table**

```
REVOKE ALL PRIVILEGES
ON Employees
FROM 'dept_user';
```

**d) Develop a query to revoke some privileges of employees table from departments table**

```
REVOKE INSERT
ON Employees
FROM 'dept_user';
```

e) **Write a query to implement the save point.**

```
-- Start transaction
START TRANSACTION;

-- First operation
UPDATE Employees
SET emp_salary = emp_salary + 1000
WHERE dept_no = 1;

-- Create savepoint
SAVEPOINT sp1;

-- Second operation
UPDATE Employees
SET emp_salary = emp_salary + 2000
WHERE dept_no = 2;

-- Rollback to savepoint (undo second update only)
ROLLBACK TO sp1;

-- Commit final changes
COMMIT;
```

**FULL CODE :**

```
-- 1. CREATE DATABASE
CREATE DATABASE CompanyDB;

-- 2. USE DATABASE
USE CompanyDB;


-- 3. CREATE TABLES

CREATE TABLE Departments (
  dept_no INT PRIMARY KEY,
  dept_name VARCHAR(50),
  dept_location VARCHAR(50)
);

CREATE TABLE Employees (
```

```sql
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  emp_salary INT,
  dept_no INT,
  FOREIGN KEY (dept_no) REFERENCES Departments(dept_no)
);


-- 4. INSERT SAMPLE DATA

INSERT INTO Departments VALUES
(1, 'HR', 'Delhi'),
(2, 'Finance', 'Mumbai'),
(3, 'IT', 'Bangalore');

INSERT INTO Employees VALUES
(101, 'Amit', 30000, 1),
(102, 'Riya', 35000, 2),
(103, 'Rahul', 40000, 3),
(104, 'Sneha', 32000, 1),
(105, 'Karan', 37000, 2);


-- 5. CREATE USER (for privilege operations)
CREATE USER 'dept_user' IDENTIFIED BY '1234';


-- (a) GRANT ALL PRIVILEGES
GRANT ALL PRIVILEGES
ON Employees
TO 'dept_user';


-- (b) GRANT SOME PRIVILEGES
GRANT SELECT, INSERT
ON Employees
TO 'dept_user';


-- (c) REVOKE ALL PRIVILEGES
REVOKE ALL PRIVILEGES
ON Employees
FROM 'dept_user';


-- (d) REVOKE SOME PRIVILEGES
REVOKE INSERT
ON Employees
```

```sql
FROM 'dept_user';


-- (e) SAVEPOINT IMPLEMENTATION

START TRANSACTION;

UPDATE Employees
SET emp_salary = emp_salary + 1000
WHERE dept_no = 1;

SAVEPOINT sp1;

UPDATE Employees
SET emp_salary = emp_salary + 2000
WHERE dept_no = 2;

ROLLBACK TO sp1;

COMMIT;
CREATE USER 'dept_user'@'localhost' IDENTIFIED BY '1234';

-- FINAL CHECK
SELECT * FROM Employees;
```

**OUTPUT :**

Log | 1: Employees [5]

In **DbVisualizer Pro** this feature includes data editing functionality, management of binary/BLOB data, and more...

| * | emp_id | emp_name | emp_salary | dept_no |
|---|---|---|---|---|
| 1 | 101 | Amit | 31000 | 1 |
| 2 | 102 | Riya | 35000 | 2 |
| 3 | 103 | Rahul | 40000 | 3 |
| 4 | 104 | Sneha | 33000 | 1 |
| 5 | 105 | Karan | 37000 | 2 |